3 (a) Draw **one** line to connect each **Operating System (OS)** term to the **most appropriate** description about it.

OS term	**Description**
Multi-tasking	Using secondary storage to simulate additional main memory
Paging	Managing the processes running on the CPU
Interrupt handling	Managing the execution of many programs that appear to run at the same time
Scheduling	Locating non-contiguous blocks of data and relocating them
Virtual memory	Transferring control to another routine when a service is required
	Reading/writing same-size blocks of data from/to secondary storage when required

[5]

(b) Explain how an interpreter executes a program without producing a complete translated version of it.

[4]

4 **(a)** **(i)** Explain why Reverse Polish Notation (RPN) is used to carry out the evaluation of expressions.

....

....

....

.... [2]

(ii) Identify, with reasons, a data structure that could be used to evaluate an expression in RPN.

....

....

....

.... [2]

(b) Write the infix expression in RPN.

```
(a – b) * (a + c) / 7
```

....

.... [1]

(c) Write the RPN expression as an infix expression.

```
a b / 4 * a b + -
```

....

.... [1]

(d) Evaluate the RPN expression:

```
a b + c d / /
```

where `a = 17, b = 3, c = 48 and d = 12`.

Show your working.

....

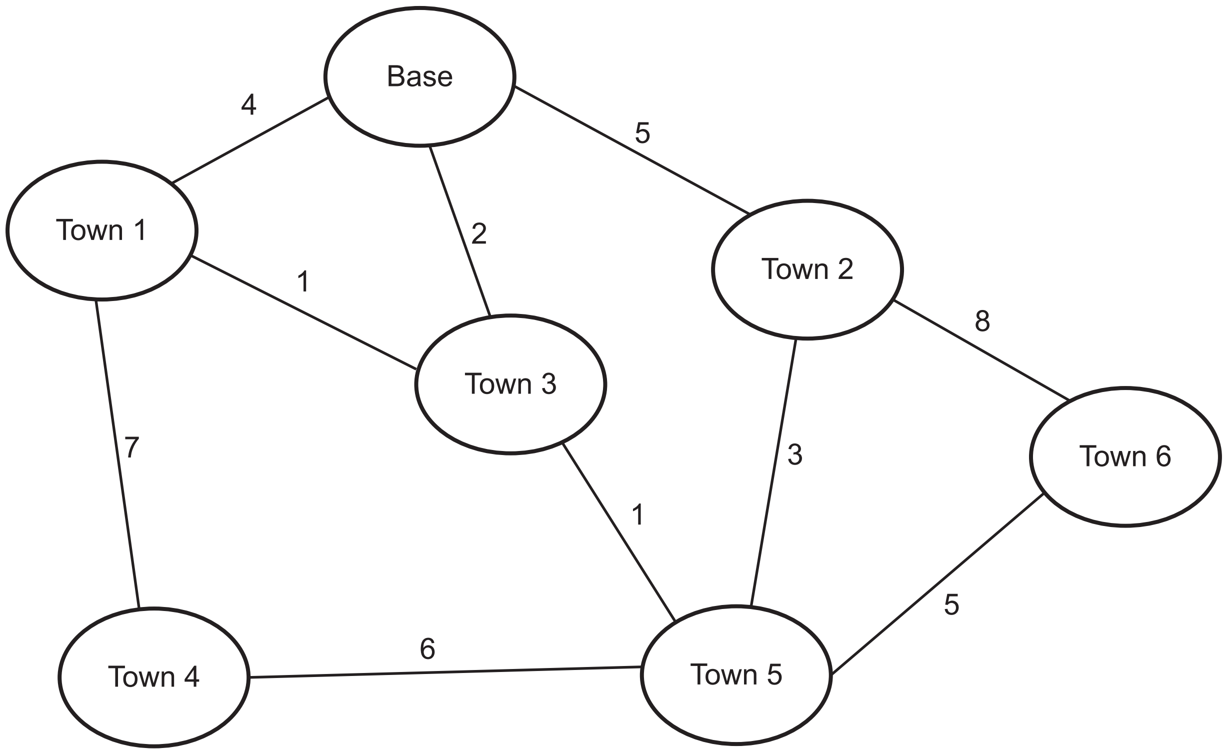
....

....

.... [2]

5 (a) Calculate the shortest distance between the base and each of the other towns in the diagram using Dijkstra's algorithm.

Show your working **and** write your answers in the table provided.



Working ....

....

....

....

....

....

....

....

Answers

Town 1	**Town 2**	**Town 3**	**Town 4**	**Town 5**	**Town 6**

[5]

(b) Explain the use of graphs to aid Artificial Intelligence (AI).

....

....

....

....

....

.... [3]

6 Give **two** benefits **and two** drawbacks of packet switching.

Benefit 1 ....

....

Benefit 2 ....

....

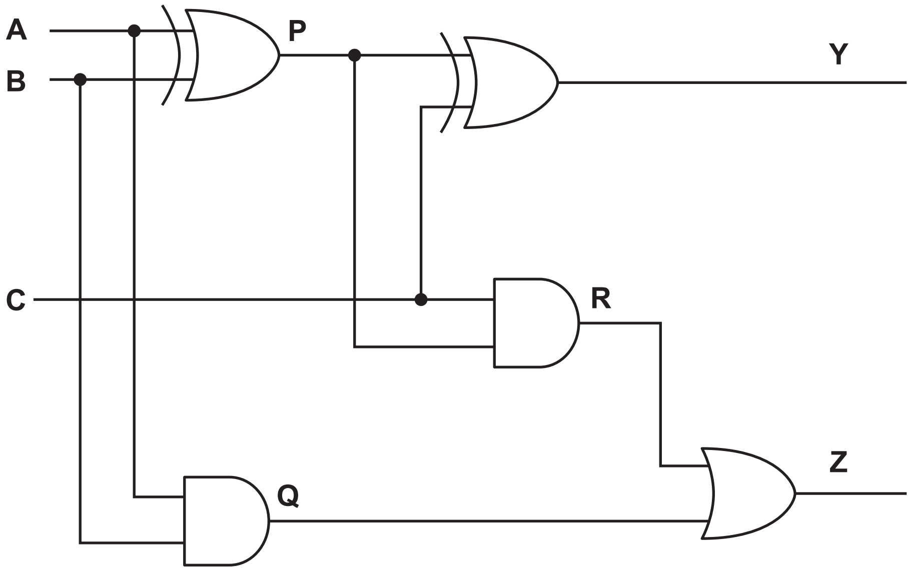
Drawback 1 ....

....

Drawback 2 ....

....
[4]

7 The diagram shows a logic circuit.



(a) Complete the truth table for the given logic circuit. Show your working.

Inputs			Working space			Outputs	
A	**B**	**C**	**P**	**Q**	**R**	**Y**	**Z**
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

[3]

(b) State the name of the logic circuit.

.... [1]

(c) Write the Boolean expressions for the two outputs **Y** and **Z** in the truth table as sum-of-products **and** state the purpose of each output.

Y = ...

Purpose ....

Z = ...

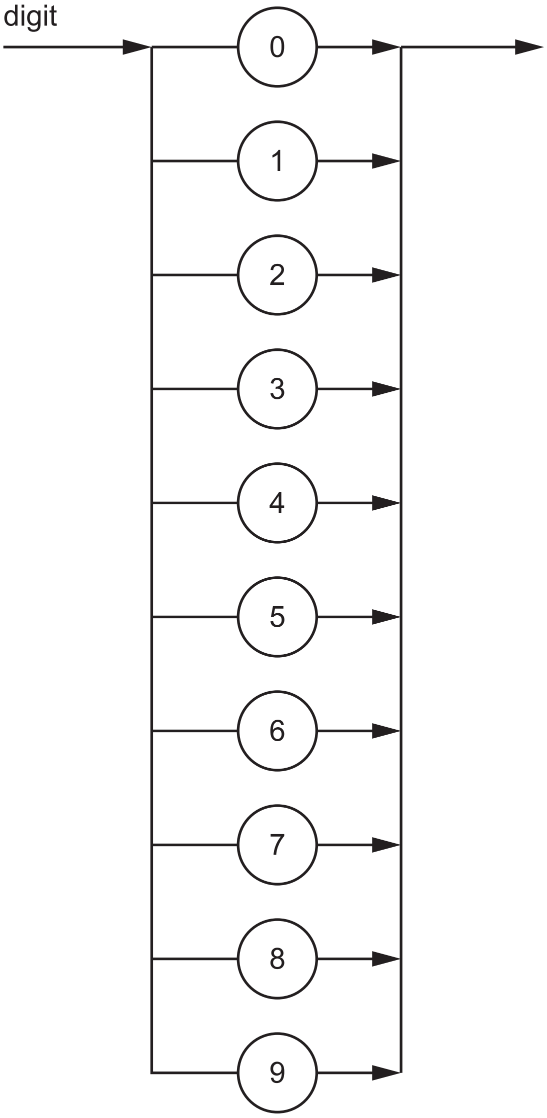
Purpose ....

[4]

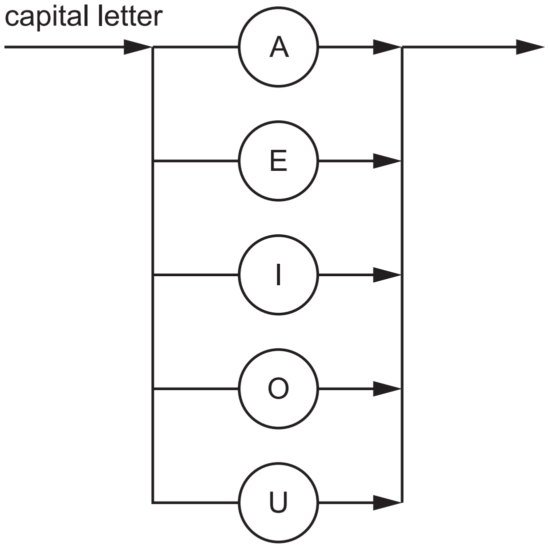
4 The following syntax diagrams for a particular programming language show the syntax of:

- a digit
- a capital letter
- a character.

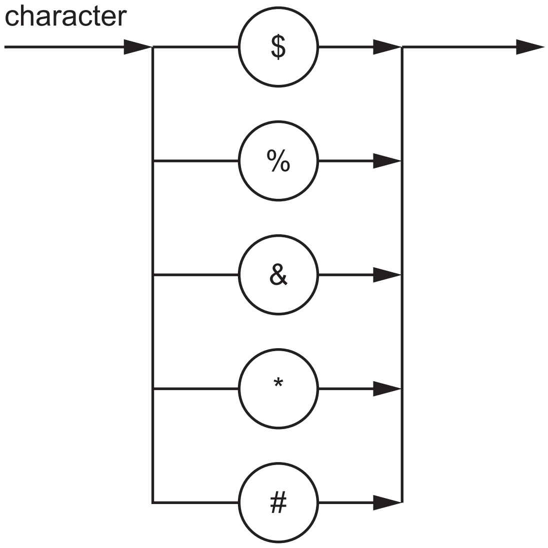
digit



capital letter



character



(a) Write the Backus-Naur Form (BNF) notation of the syntax diagram for character.

....

....

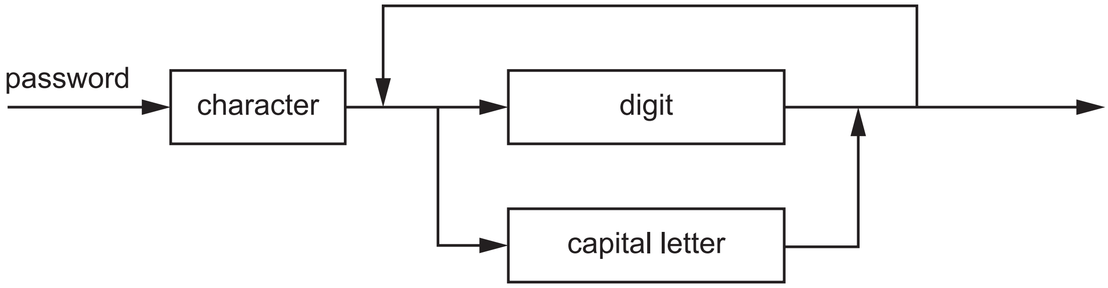
.... [2]

(b) A password must begin with a character and be followed by one or more digits or capital letters.

(i) State an example of a valid password.

.... [1]

(ii) A valid password is represented by the syntax diagram:



Write the BNF notation of the syntax diagram for password.

....

....

....

....

....

....

....

.... [4]

6 **(a)** Explain how packet switching is used to transfer messages across the internet.

....

....

....

....

....

....

....

....

....

.... [5]

(b) Outline the function of a router in packet switching.

....

....

....

....

....

.... [3]

7 **(a)** Write the Boolean expression that corresponds to the given truth table as a sum-of-products.

INPUT				**OUTPUT**
A	**B**	**C**	**D**	**Z**
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Z = ...

.... [3]

(b) **(i)** Complete the Karnaugh map (K-map) for the given truth table.

CD \ AB	00	01	11	10
00				
01				
11				
10				

[2]

(ii) Draw loop(s) around appropriate group(s) of 1s in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to **part b(ii)** as a simplified sum-of-products.

Z = ...

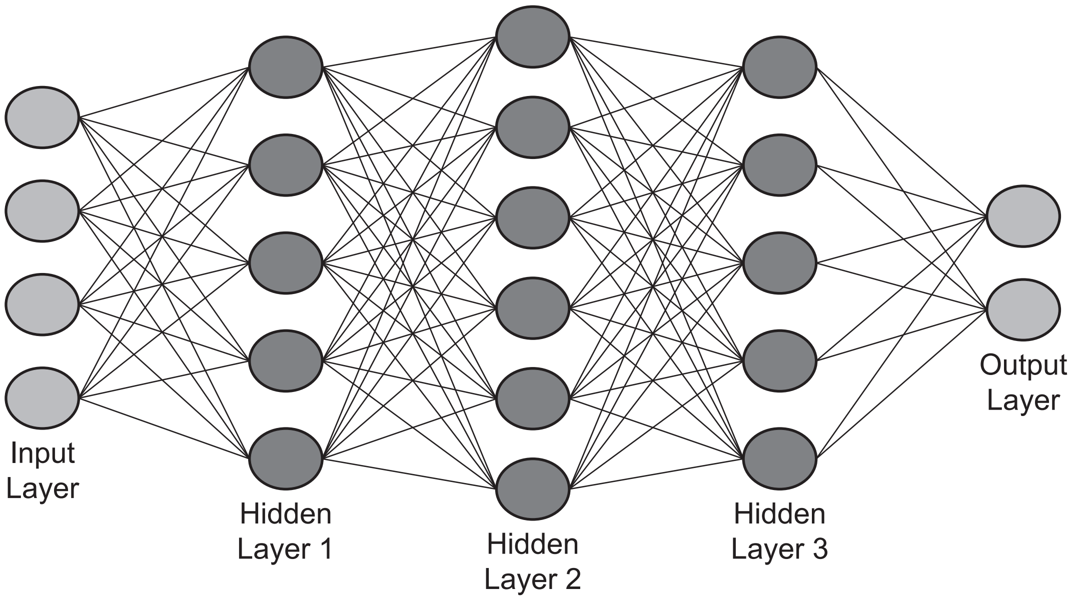
.... [2]

(iv) Write the simplified Boolean expression for your answer to **part b(iii)**.

Z = ...

.... [1]

9 **(a)** The diagram shown represents an artificial neural network.



(i) State the reason for having multiple hidden layers in an artificial neural network.

....

.... [1]

(ii) Explain how artificial neural networks enable machine learning.

....

....

....

....

....

....

....

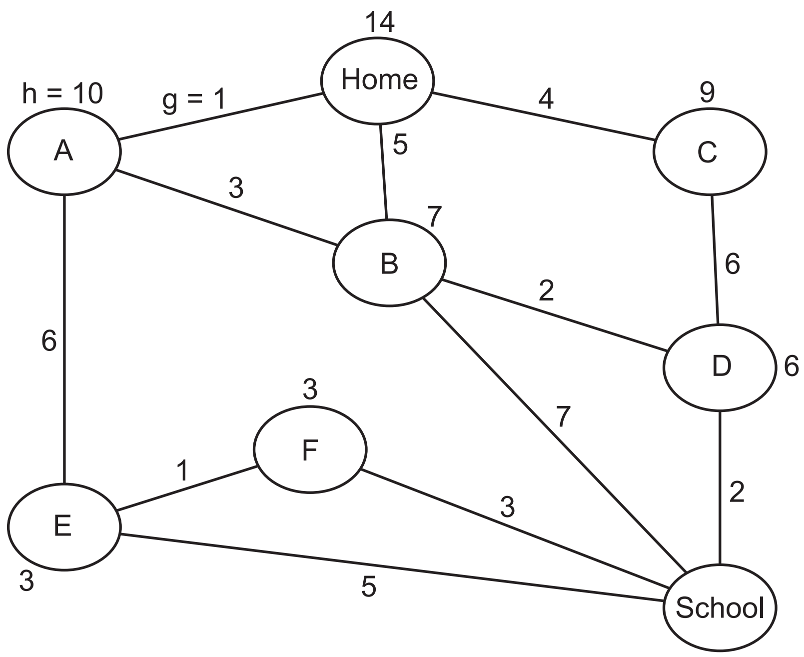
....

....

.... [4]

(b) Find the shortest path between the Home and School nodes using the A* algorithm. Show your working in the table provided.

The first two rows in the table have been completed.



Node	Cost from Home node (g)	Heuristic (h)	Total (f = g + h)
Home	0	14	14
A	1	10	11

Final path	

[5]

3 Data can be sent over networks using either circuit switching or packet switching.

Describe both methods of data transmission. Include a different advantage and disadvantage for each method.

Circuit switching ....

....

....

....

....

Advantage ....

....

Disadvantage ....

....

Packet switching ....

....

....

....

....

Advantage ....

....

Disadvantage ....

....
[8]

4 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

(a) Describe what is meant by **RISC** and **CISC processors**.

RISC ....

....

....

....

CISC ....

....

....

....

[4]

(b) Identify **two** differences between RISC and CISC processors.

1 ....

....

....

2 ....

....

....

[2]

5 Part of a program's calculations uses the integer variables `j`, `k`, `m`, `n` and `p`.

```
j = 3
k = 2
m = 10
n = (j + k)/(j - k)
p = m * (m - j * k)
```

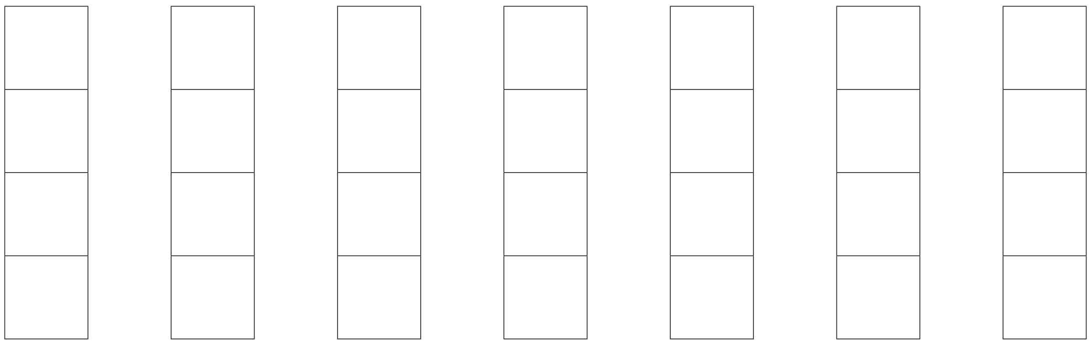
(a) Write the Reverse Polish Notation (RPN) for the expression:

```
(j + k)/(j - k)
```

.... [2]

(b) **(i)** Show the changing contents of the stack as the value for `p` is calculated from its RPN expression:

```
m m j k * - *
```



[4]

(ii) Describe the main steps in the evaluation of this RPN expression using a stack.

....

....

....

....

....

....

....

.... [4]

(c) State **two other** uses of a stack.

1 ....

....

2 ....

....
[2]

6 A virtual machine is used to emulate a new computer system.

Describe **two** benefits and **one** limitation of using a virtual machine for this purpose.

Benefit 1 ....

....

....

....

....

Benefit 2 ....

....

....

....

....

Limitation ....

....

....

....

....
[6]

9 The table shows assembly language instructions for a processor that has one general purpose register, the Accumulator (ACC).

	Instruction		Explanation
Label	**Opcode**	**Operand**	
	`LDM`	`#n`	Load the number n to ACC
	`LDD`	`<address>`	Load the contents of the given address to ACC
	`LDI`	`<address>`	The address to be used is at the given address Load the contents of this second address to ACC
	`ADD`	`<address>`	Add the contents of the given address to the ACC
	`STO`	`<address>`	Store the contents of the ACC at the given address
`<label>:`		`<data>`	Gives a symbolic address <label> to the memory location with the contents <data> <label> can be used in place of <address>
# denotes a denary number, e.g. #123			

(a) The address `500` contains the value `100` and the address `100` contains the value `20`.

State the addressing mode and the contents of ACC after each instruction has been executed.

`LDM #500` Addressing mode ....

Contents of ACC ....

`LDD 500` Addressing mode ....

Contents of ACC ....

`LDI 500` Addressing mode ....

Contents of ACC ....

[3]

(b) Use only the given instruction set to write **assembly language** code to:

- use the constant `20` which needs to be stored
- add this constant to the value stored in the address contained in the variable `Y`
- store the result in variable `Z`.

Label	Instruction	
	Opcode	**Operand**

[7]

(c) Write the goal, using the variable `X`, to find all the students who have a tutor that teaches them. For example, Hua has Alan for a tutor and is also taught mathematics by Alan.

....

....

....

.... [4]

3 The TCP/IP protocol suite has four layers. The application layer provides user services.

(a) Identify **two** protocols used by this layer. Describe the use of each protocol.

Protocol 1 ....

Description ....

....

....

Protocol 2 ....

Description ....

....

....
[4]

(b) Identify **two other** layers of the TCP/IP protocol suite. Describe the function of each layer.

Layer 1 ....

Description ....

....

....

Layer 2 ....

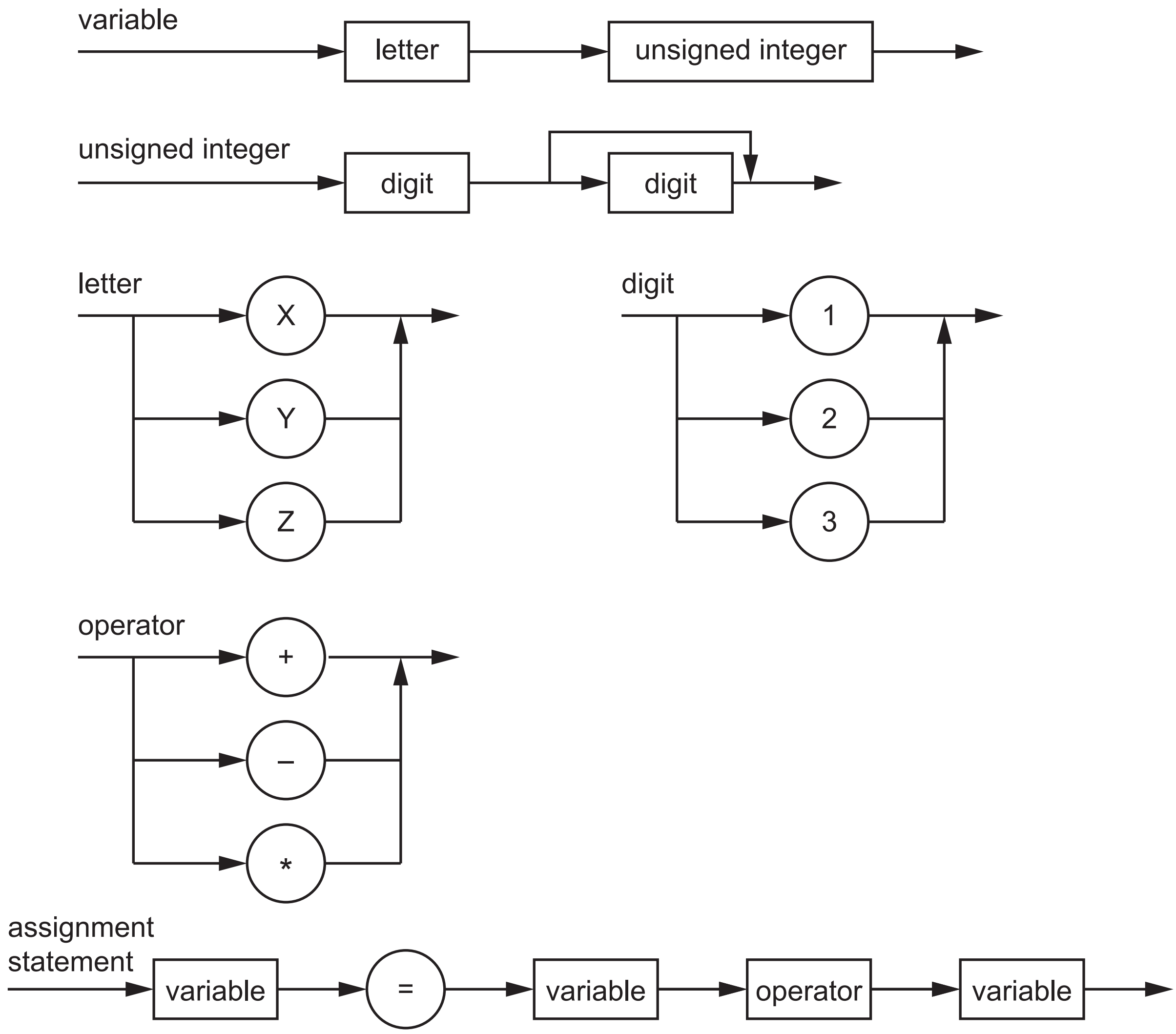
Description ....

....

....
[4]

4 The following syntax diagrams show the syntax of:

- a variable
- an unsigned integer
- a letter
- a digit
- an operator
- an assignment statement.



(a) The following assignment statements are invalid. State the reason in each case.

```
X1 = Y2 - 12
```

Reason ....

....

```
Z = Y12 + Z1
```

Reason ....

....

[2]

(b) Complete the Backus-Naur Form (BNF) for the syntax diagrams shown.

`<letter>` has been completed for you.

`<variable> ::=` ....

....

`<unsigned_integer> ::=` ....

....

`<letter> ::= X | Y | Z`

`<digit> ::=` ....

....

`<operator> ::=` ....

....

`<assignment_statement> ::=` ....

.... [5]

(c) The syntax of an assignment statement is changed to allow each of the variables on the right-hand side of the '=' symbol to be either a variable or an unsigned integer.

(i) Draw a syntax diagram for the new syntax of the **assignment statement**.

[3]

(ii) Write the Backus-Naur Form (BNF) for your syntax diagram.

....

....

....

.... [3]

5 There are four basic categories of computer architecture. Single Instruction Single Data (SISD) is one architecture.

Identify the **three other** categories of computer architecture.

Describe each category that you identify.

Architecture 1 ....

Description ....

....

....

....

Architecture 2 ....

Description ....

....

....

....

Architecture 3 ....

Description ....

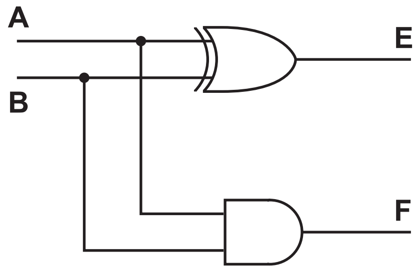
....

....

....

[6]

6 A logic circuit has two inputs **A** and **B**, and two outputs **E** and **F**.



(a) Complete the truth table for this logic circuit.

INPUT		**OUTPUT**	
A	**B**	**E**	**F**
0	0		
0	1		
1	0		
1	1		

[2]

(b) **(i)** State the name of this logic circuit.

.... [1]

(ii) State the purpose of each output **E** and **F**.

Purpose of **E** ....

Purpose of **F** ....

[2]

5 (a) Write the infix expression in Reverse Polish Notation (RPN).

```
a * b + b - d + 15
```

....

.... [1]

(b) (i) Write the RPN expression in infix form.

```
a b - c d + * a /
```

....

.... [1]

(ii) Evaluate your infix expression from **part (b)(i)** when a = 5, b = 10, c = 27 and d = 12.

....

.... [1]

6 A message is encrypted using a private key and sent to an individual using asymmetric encryption.

(a) State what is meant by a **private key**.

....

....

....

.... [2]

(b) Describe the process of asymmetric encryption.

....

....

....

.... [2]

(c) Explain how a digital signature is used to verify a message when it is received.

....

....

....

....

....

....

....

.... [4]

7 **(a)** Complete the Karnaugh map (K-map) for the Boolean expression.

$$\mathbf{Z = \overline{A}.\ B.\ \overline{C}.\ \overline{D} + \overline{A}.\ B.\ \overline{C}.\ D + A.\ B.\ \overline{C}.\ \overline{D} + A.\ B.\ \overline{C}.\ D + A.\ \overline{B}.\ \overline{C}.\ \overline{D} + A.\ \overline{B}.\ \overline{C}.\ D}$$

CD \ AB	**00**	**01**	**11**	**10**
00				
01				
11				
10				

[2]

(b) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(c) Write the Boolean expression from your answer to **part (b)** as a simplified sum-of-products. Use Boolean algebra to give your answer in its simplest form.

Simplified sum-of-products

Z = ...

Simplest form

Z = ...

[3]

8 Virtual memory, paging and segmentation are used in memory management.

(a) Explain what is meant by **virtual memory**.

....

....

....

....

....

.... [3]

(b) State **one** difference between paging and segmentation in the way memory is divided.

....

.... [1]

9 Deep learning is used in Artificial Intelligence (AI).

(a) Describe what is meant by **deep learning**.

....

....

....

.... [2]

(b) Outline the reasons for using deep learning.

....

....

....

.... [2]

10 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

(a) Tick (✓) **one** box in each row to show if the statement applies to RISC or CISC processors.

Statement	RISC	CISC
uses a smaller instruction set		
uses single-cycle instructions and limited addressing modes		
uses fewer general-purpose registers		
uses both hardwired and micro-coded control unit		
uses a system where cache is split between data and instructions		

[2]

(b) Describe the process of pipelining during the fetch-execute cycle in RISC processors.

....

....

....

....

....

....

....

.... [4]

2 Lexical analysis and syntax analysis are stages in the compilation of a program.

(a) Identify **two other** stages that take place during the compilation of a program.

1 ....

....

2 ....

.... [2]

(b) Outline the purpose of syntax analysis.

....

....

....

.... [2]

3 **(a)** Explain why a protocol is used in communication between computers.

....

....

....

.... [2]

(b) The TCP/IP protocol implementation can be viewed as a stack.

Complete the diagram for the TCP/IP protocol stack.

Transport
Link

[2]

(c) Describe the purpose of the IMAP protocol.

....

....

....

.... [2]

(ii) Write **pseudocode** to store the details of the following flight in the variable you set up in **part (c)(i)**.

Field	Data
flight number	XA782
destination	Cambridge
date of departure	12/12/2022
type of aircraft used	C350

Use the field names you created in **part (b)**.

....

....

....

....

....

.... [3]

5 Describe what is meant by a **virtual machine**.
Include in your answer **two** benefits and **two** drawbacks of using a virtual machine.

Description ....

....

....

....

Benefit 1 ....

....

Benefit 2 ....

....

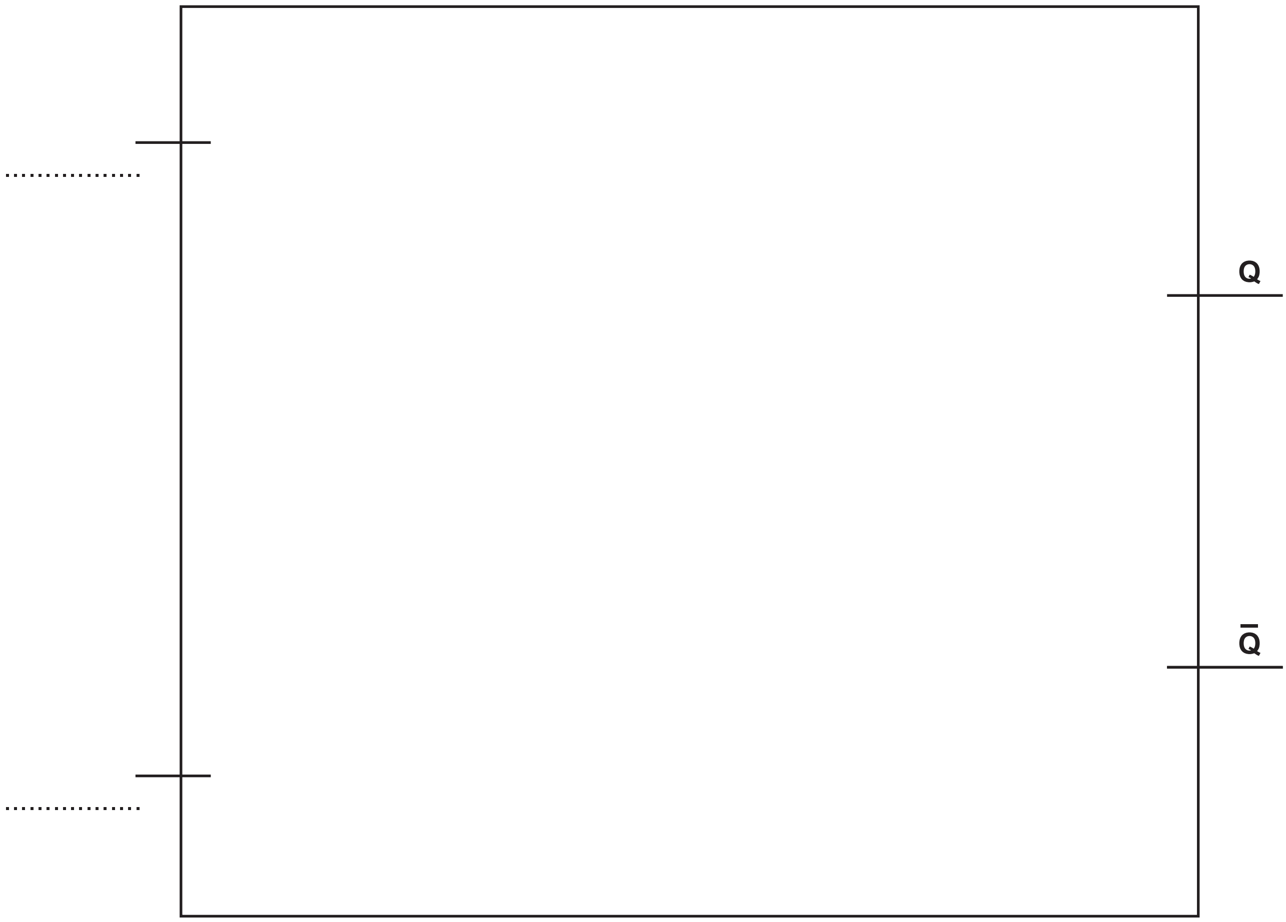
Drawback 1 ....

....

Drawback 2 ....

....
[6]

8 **(a)** Draw a logic circuit for an SR flip-flop **and** label the inputs.



[4]

(b) State the purpose of a flip-flop.

....

.... [1]

(c) Simplify the following expression using Boolean algebra, including De Morgan's laws.
Show your working.

$$\overline{\overline{(\bar{A}.\bar{B})}.\overline{(\bar{A}.C)}.\overline{(B.D)}}$$

Working ....

....

....

....

....

Answer ....

[3]

9 **(a)** Explain the need for scheduling in process management.

....

....

....

....

....

.... [3]

(b) Describe these scheduling routines **and** identify a benefit for each one.

Shortest job first ....

....

....

....

Round robin ....

....

....

....

First come first served ....

....

....

....

[6]

5 **(a)** Write the infix expression in Reverse Polish Notation (RPN).

```
a * b + b - d + 15
```

....

.... [1]

(b) **(i)** Write the RPN expression in infix form.

```
a b - c d + * a /
```

....

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(ii) Evaluate your infix expression from **part (b)(i)** when a = 5, b = 10, c = 27 and d = 12.

....

.... [1]

6 A message is encrypted using a private key and sent to an individual using asymmetric encryption.

(a) State what is meant by a **private key**.

....

....

....

.... [2]

(b) Describe the process of asymmetric encryption.

....

....

....

.... [2]

(c) Explain how a digital signature is used to verify a message when it is received.

....

....

....

....

....

....

....

.... [4]

7 **(a)** Complete the Karnaugh map (K-map) for the Boolean expression.

$$\mathbf{Z = \overline{A}.\ B.\ \overline{C}.\ \overline{D} + \overline{A}.\ B.\ \overline{C}.\ D + A.\ B.\ \overline{C}.\ \overline{D} + A.\ B.\ \overline{C}.\ D + A.\ \overline{B}.\ \overline{C}.\ \overline{D} + A.\ \overline{B}.\ \overline{C}.\ D}$$

CD \ AB	00	01	11	10
00				
01				
11				
10				

[2]

(b) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(c) Write the Boolean expression from your answer to **part (b)** as a simplified sum-of-products. Use Boolean algebra to give your answer in its simplest form.

Simplified sum-of-products

Z = ...

Simplest form

Z = ...

[3]

8 Virtual memory, paging and segmentation are used in memory management.

(a) Explain what is meant by **virtual memory**.

....

....

....

....

....

.... [3]

(b) State **one** difference between paging and segmentation in the way memory is divided.

....

.... [1]

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(a) Describe what is meant by **deep learning**.

....

....

....

.... [2]

(b) Outline the reasons for using deep learning.

....

....

....

.... [2]

10 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

(a) Tick (✓) **one** box in each row to show if the statement applies to RISC or CISC processors.

Statement	RISC	CISC
uses a smaller instruction set		
uses single-cycle instructions and limited addressing modes		
uses fewer general-purpose registers		
uses both hardwired and micro-coded control unit		
uses a system where cache is split between data and instructions		

[2]

(b) Describe the process of pipelining during the fetch-execute cycle in RISC processors.

....

....

....

....

....

....

....

.... [4]

4 Two descriptions of user-defined data types are given.

Give appropriate type declaration statements for each, including appropriate names.

(a) A data type to hold a set of prime numbers below 20. These prime numbers are:

2, 3, 5, 7, 11, 13, 17, 19

....

....

....

.... [2]

(b) A data type to point to a day in the week, for example Monday.

....

....

....

.... [2]

5 **(a)** State, with a reason, where it would be appropriate to use circuit switching.

....

....

....

.... [2]

(b) Give **two** benefits and **two** drawbacks of circuit switching.

Benefit 1 ....

....

Benefit 2 ....

....

Drawback 1 ....

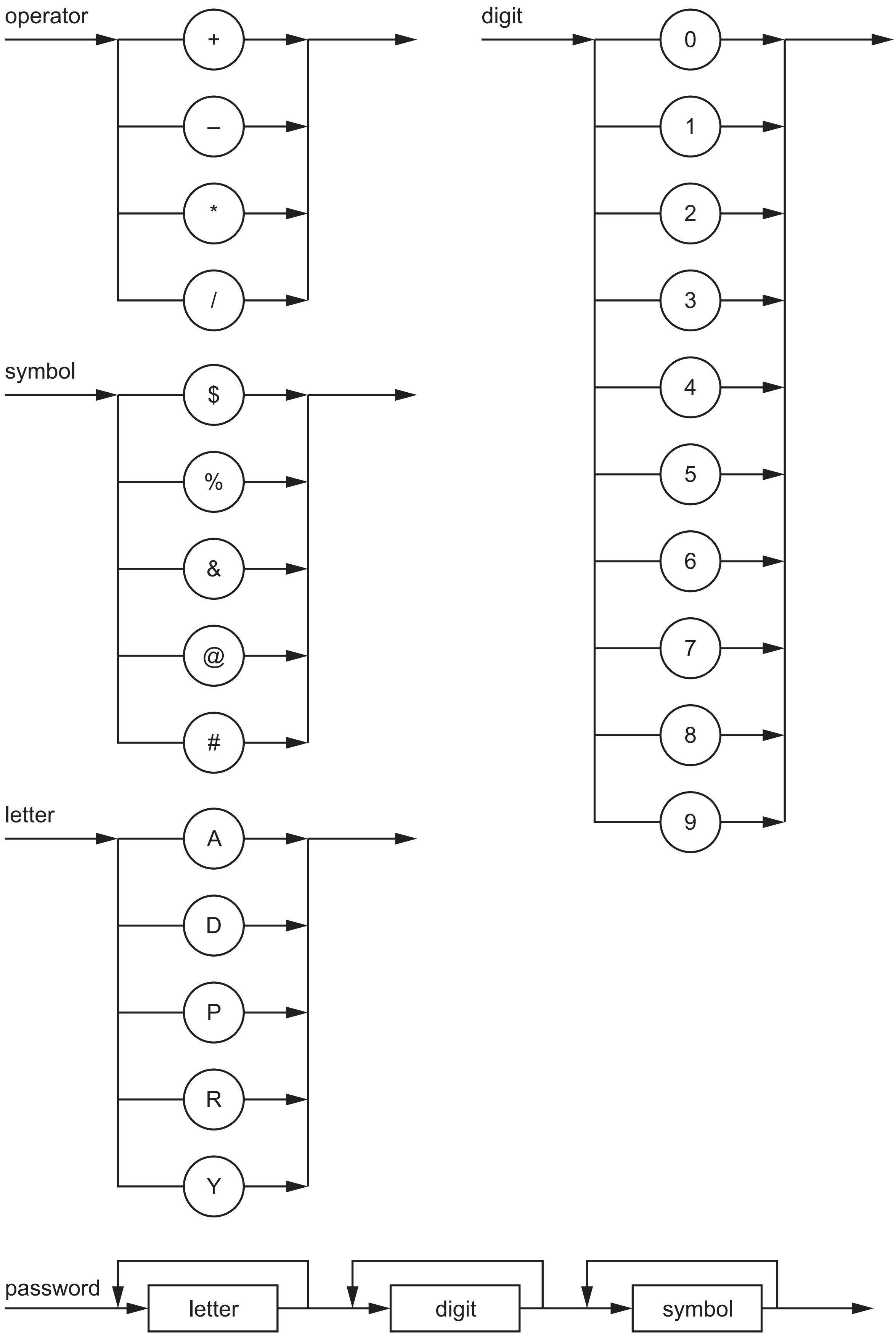
....

Drawback 2 ....

....

[4]

6 Several syntax diagrams are shown.



(a) State whether each of the following passwords is valid or invalid and give a reason for your choice.

`DPAD99$` ....

Reason ....

....

`DAD#95` ....

Reason ....

....

`ADY123?` ....

Reason ....

....

[3]

(b) Complete the Backus-Naur Form (BNF) for the syntax diagrams shown.

`<symbol> ::=` ....

....

`<letter> ::=` ....

....

[1]

(c) An identifier begins with one or more letters, followed by zero digits or one digit or more digits.

Valid letters and digits are shown in the syntax diagrams on page 6.

Draw a syntax diagram for an identifier.

[4]

7 **(a)** Complete the Karnaugh map (K-map) for the following Boolean expression.

$$\mathbf{Z = \overline{A}.\overline{B}.\overline{C}.\overline{D} + \overline{A}.\overline{B}.\overline{C}.D + \overline{A}.B.\overline{C}.\overline{D} + \overline{A}.B.\overline{C}.D + A.B.\overline{C}.\overline{D} + A.B.\overline{C}.D}$$

CD \ AB	00	01	11	10
00				
01				
11				
10				

[2]

(b) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(c) Write the Boolean logic expression from your answer to **part (b)** as a simplified sum-of-products.

Z = ...

.... [2]

(d) Use Boolean algebra to give your answer to **part (c)** in its simplest form.

Z = ... [1]

8 Outline the characteristics of massively parallel computers.

....

....

....

....

....

.... [3]

9 **(a)** Encryption is used to alter data into a form that makes it meaningless if intercepted.

Describe the purpose of asymmetric key cryptography.

....

....

....

.... [2]

(b) Identify **two** benefits and **two** drawbacks of quantum cryptography.

Benefit 1 ....

....

Benefit 2 ....

....

Drawback 1 ....

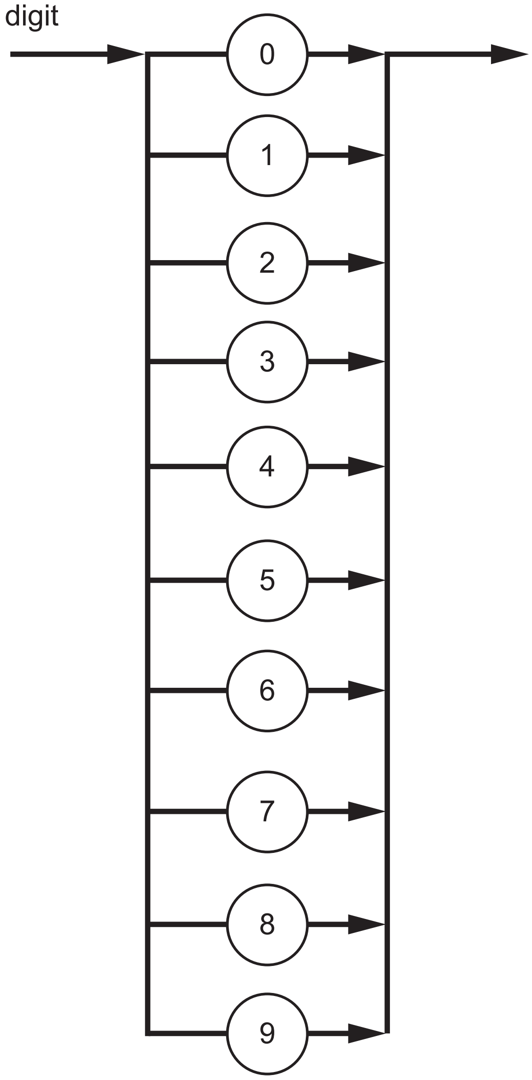
....

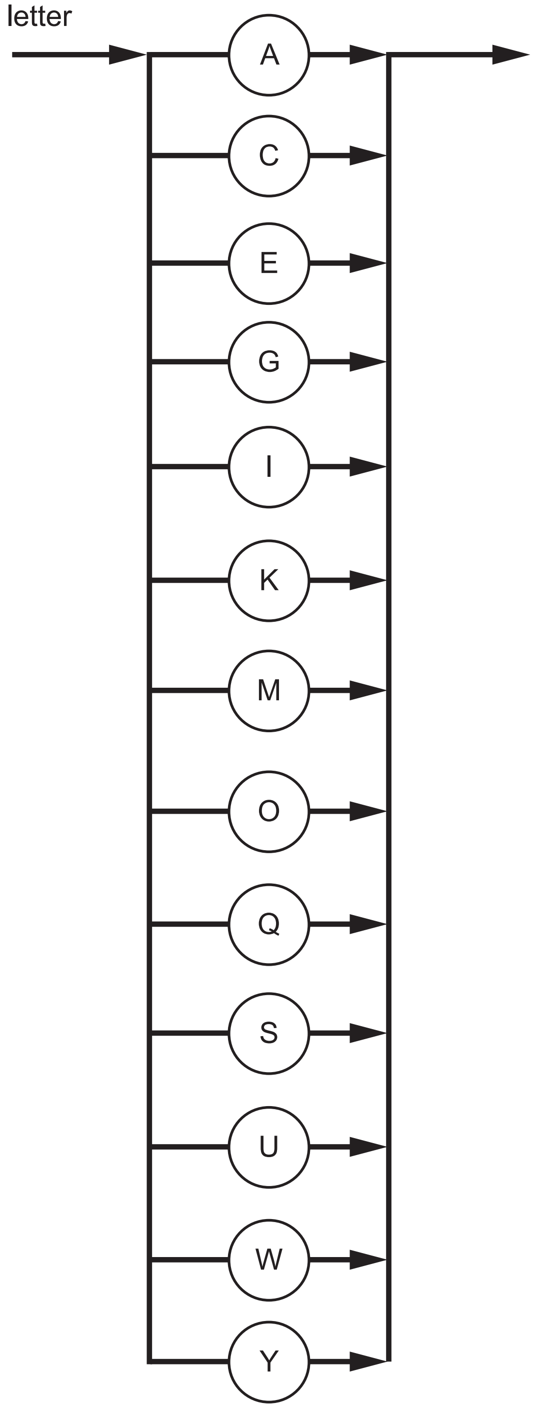
Drawback 2 ....

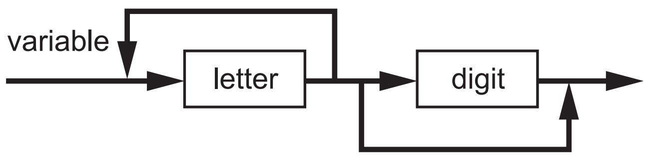
....

[4]

3 Several syntax diagrams are shown.







(a) State whether each variable is valid or invalid **and** give a reason for your choice in each case.

`9SW` ....

Reason ....

....

`UWY` ....

Reason ....

.... [2]

(b) `<word>` contains one or more letters.

Complete the Backus-Naur Form (BNF) for `<word>` **and** use this to complete the BNF for `<variable>`.

`<word> ::=` ....

....

`<variable> ::=` ....

.... [3]

(c) Vehicle registrations must begin with two letters and be followed by one, two or three digits.

Valid letters and digits are shown in the syntax diagrams on page 4.

(i) State an example of a valid vehicle registration.

....

.... [1]

(ii) Draw a syntax diagram for a vehicle registration.

[3]

7 **(a)** State **two** examples of where it would be appropriate to use packet switching.

....

....

....

.... [2]

(b) Give **four** differences between circuit switching and packet switching.

1 ....

....

2 ....

....

3 ....

....

4 ....

....

[4]

8 **(a)** Describe the use of pipelining in Reduced Instruction Set Computers (RISC).

....

....

....

.... [2]

(b) The processing of instructions is divided into five stages:

- instruction fetch (IF)
- instruction decode (ID)
- operand fetch (OF)
- instruction execute (IE)
- write back result (WB)

Each stage is carried out using a different register when pipelining is used.

Complete the table to show how a program consisting of **six** instructions would be completed using pipelining.

Clock cycles

Processor stages	1	2	3	4	5	6	7	8	9	10	11	12
IF												
ID												
OF												
IE												
WB												

[4]

9 This truth table represents a logic circuit.

INPUT				OUTPUT
A	B	C	D	Z
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

(a) Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

Z = ...

.... [3]

(b) Complete the Karnaugh map (K-map) for the given truth table.

CD \ AB	00	01	11	10
00				
01				
11				
10				

[2]

(c) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(d) Write the Boolean logic expression from your answer to **part (c)** as a simplified sum-of-products.

Z = ...

.... [2]

(e) Use Boolean algebra to give your answer to **part (d)** in its simplest form.

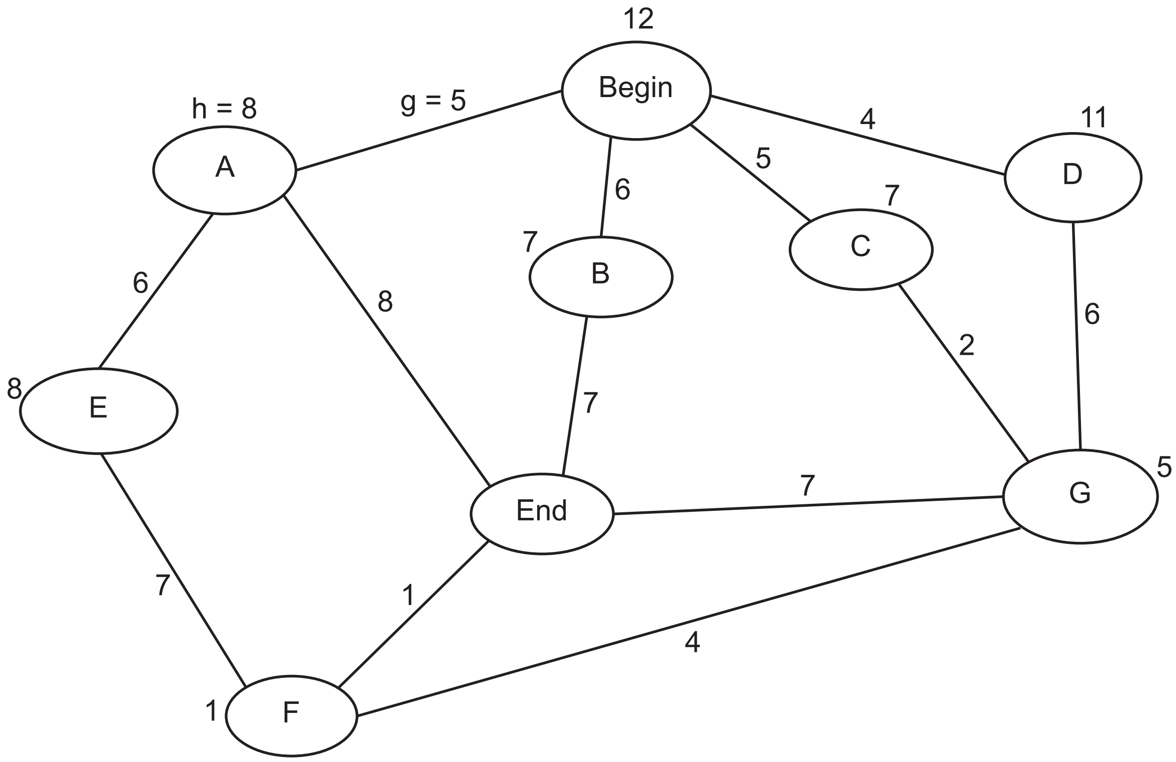
Z = ... [1]

10 (a) State **one** category of machine learning.

.... [1]

(b) Calculate the path that takes the shortest time to travel from the Begin node to the End node, using the A* algorithm.
Show your working in the table provided.

The first two rows have already been completed.



Start node	Destination node	Cost from start node (g)	Heuristic (h)	Total (f = g + h)
Begin	Begin	0	12	12
Begin	A	5	8	13

Final path	

[5]

2 (a) Draw **one** line to connect **each** protocol to its most appropriate use.

Protocol	**Use**
HTTP	to provide peer-to-peer file sharing
BitTorrent	when retrieving email messages from a mail server over a TCP/IP connection
SMTP	when transmitting hypertext documents
IMAP	to map MAC addresses onto IP addresses
	when sending email messages towards the intended destination

[4]

(b) Outline the purpose of the **Link layer** in the TCP / IP protocol suite.

....

....

....

.... [2]

3 Describe what is meant by **enumerated** and **pointer** data types.

Enumerated ....

....

....

....

Pointer ....

....

....

....

[4]

4 **(a)** Describe sequential and random methods of file organisation.

Sequential file organisation ....

....

....

....

....

Random file organisation ....

....

....

....

.... [4]

(b) Outline the process of sequential access for serial and sequential files.

....

....

....

.... [2]

5 Describe the features of SISD and MIMD computer architectures.

SISD ....

....

....

....

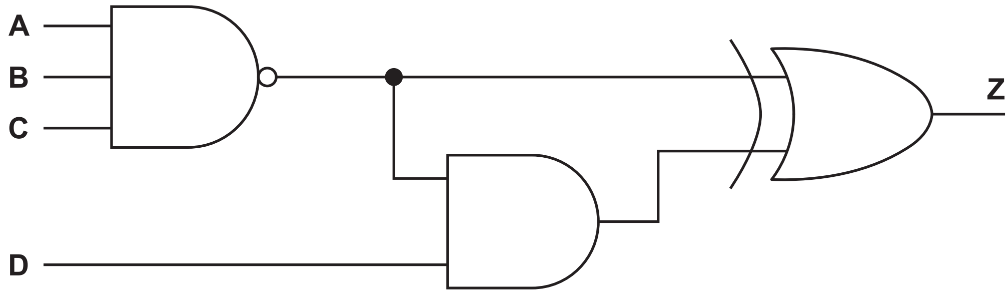
MIMD ....

....

....

.... [4]

6 This diagram represents a logic circuit.



(a) Complete the truth table for the given logic circuit.

A	B	C	D	Working space	Z
0	0	0	0		
0	0	0	1		
0	0	1	0		
0	0	1	1		
0	1	0	0		
0	1	0	1		
0	1	1	0		
0	1	1	1		
1	0	0	0		
1	0	0	1		
1	0	1	0		
1	0	1	1		
1	1	0	0		
1	1	0	1		
1	1	1	0		
1	1	1	1		

[3]

(b) Simplify the given Boolean expression using Boolean algebra.
Show your working.

$$\mathbf{Y = \overline{A}.\overline{B}.\overline{C}.\overline{D} + \overline{A}.\overline{B}.C.\overline{D} + \overline{A}.B.\overline{C}.\overline{D} + \overline{A}.B.C.\overline{D}}$$

....

....

....

....

....

.... [3]

7 **(a)** A student buys a new computer.

State **one** benefit to the student of a user interface **and** give an example.

Benefit ....

....

....

Example ....

....
[2]

(b) Two of the process states are the running state and the ready state.

Identify **one other** process state.

.... [1]

(c) Outline conditions under which a process could change from the running state to the ready state.

....

....

....

.... [2]

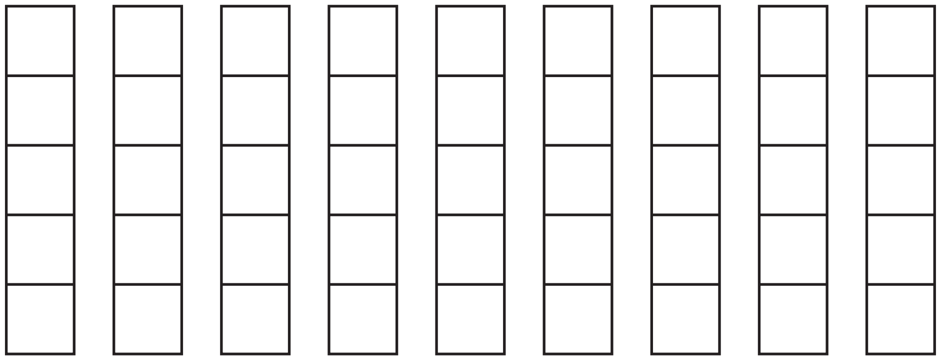
9 **(a)** **(i)** Write the infix expression for this Reverse Polish Notation (RPN) expression:

```
5 2 − 5 4 + * 9 /
```

....

.... [2]

(ii) Show how the contents of the following stack will change as the RPN expression in part **(a)(i)** is evaluated.



[4]

(b) Explain how a stack can be used to evaluate RPN expressions.

....

....

....

....

....

.... [3]

4 Complete the following paragraph about a **protocol suite**, using words from the given list.

Some words are **not** used.

BitTorrent	**circuit switching**	**layered**	**link**	**list**
peer-to-peer	**queue**	**stack**	**star**	**TCP/IP**

The protocols in a .... determine the interconnectivity rules for a .... network model such as the .... model.

[3]

5 **(a)** Outline the reasons why an operating system may need to use virtual memory.

....

....

....

.... [2]

(b) Explain the circumstances in which disk thrashing could occur.

....

....

....

....

....

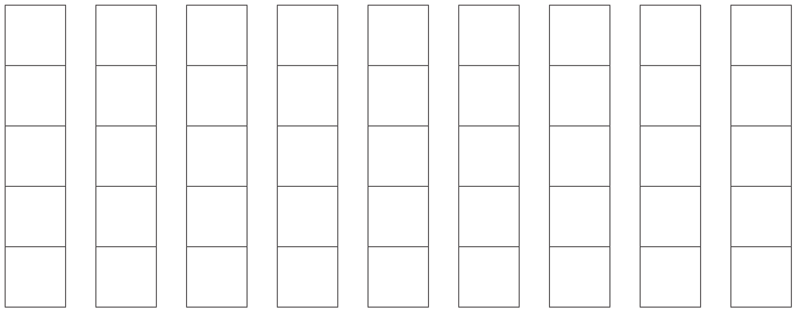
.... [3]

6 **(a)** The Reverse Polish Notation (RPN) expression:

a b * 2 / c d / *

is to be evaluated where a = 20, b = 3, c = 10 and d = 5.

Show the changing contents of the following stack as the RPN expression is evaluated.



[4]

(b) Explain how an expression stored in RPN can be evaluated.

....

....

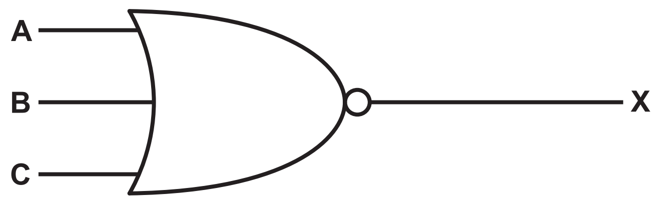
....

....

....

.... [3]

7 **(a)** This logic circuit represents the Boolean expression: $\mathbf{X = \overline{A + B + C}}$



Complete this truth table for the given logic circuit.

A	B	C	X
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

[1]

(b) Apply De Morgan's laws to the expression: $\mathbf{X = \overline{A + B + C}}$

X = ... [1]

(c) Simplify the following expression using Boolean algebra.

Show all the stages in your simplification.

$\mathbf{T = X.Y.Z + X.\overline{Y}.Z + \overline{X}}$

....

....

....

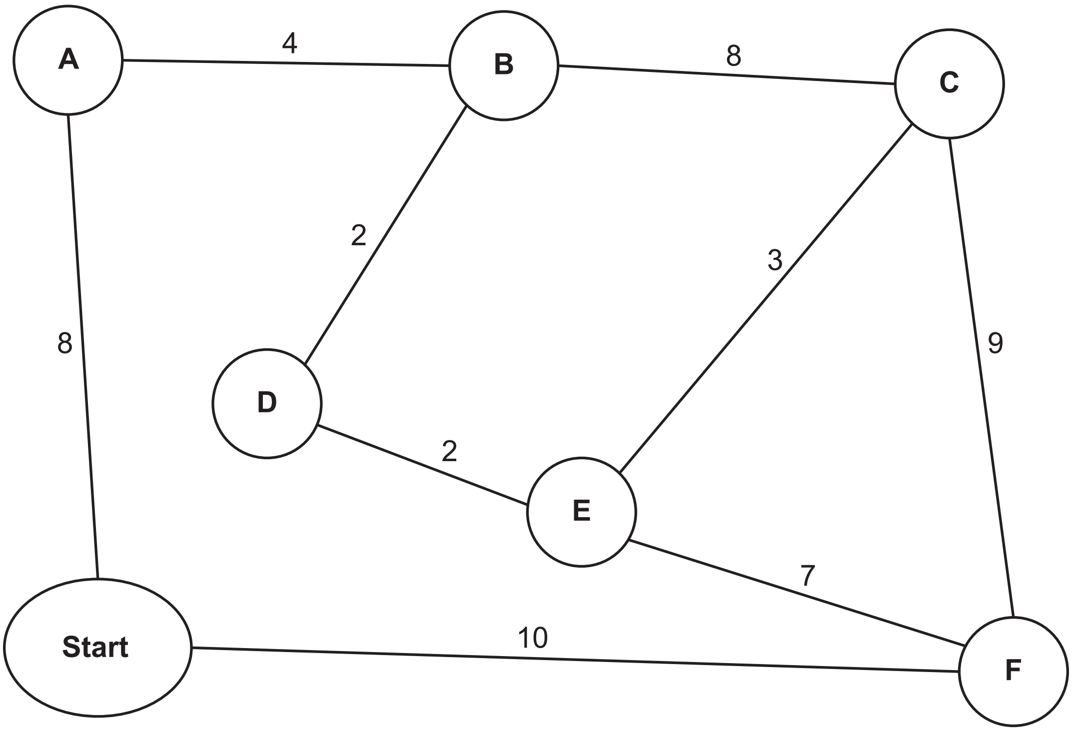
....

....

.... [3]

8 Calculate the shortest distance between the **Start** and each of the destinations in the diagram using Dijkstra's algorithm.

Show your working **and** write your answers in the table provided.



Working ....

....

....

....

....

....

Answers:

A	B	C	D	E	F

[5]

(b) Justify the use of a linked list instead of an array to implement a stack.

....

....

....

.... [2]

(c) Explain how a compiler makes use of a stack when translating recursive programming code.

....

....

....

....

....

....

....

.... [4]

10 Describe the features of the SIMD and MISD computer architectures.

SIMD ....

....

....

....

MISD ....

....

....

....

[4]

2 The TCP / IP protocol suite has four layers:

Transport, Application, Link, Internet

(a) Complete the diagram to show the correct order for these layers.

[2]

(b) Describe the function of the Transport layer.

....

....

....

.... [2]

(c) Outline **one** protocol that is associated with the Application layer.

....

....

....

.... [2]

3 **(a)** Explain what is meant by **non-composite** and **composite** data types.

....

....

....

....

....

.... [3]

5 **(a)** Write this Reverse Polish Notation (RPN) in infix form:

```
5 2 + 9 3 - / 3 *
```

[3]

(b) Write this infix expression in RPN:

```
((7 + 3) - (2 * 8)) / 6
```

[2]

(c) Evaluate this RPN expression:

```
a b - c d + * e /
```

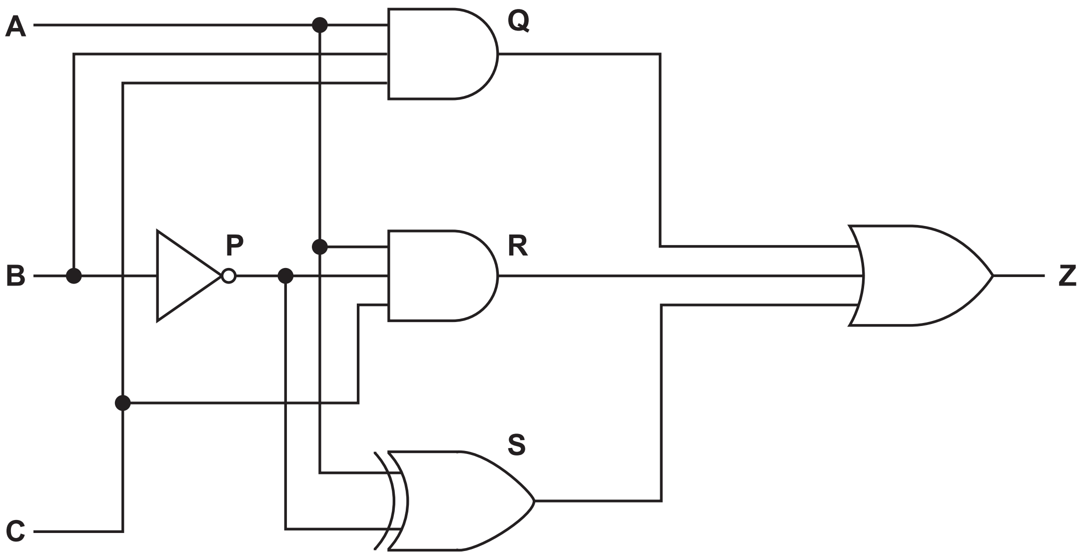
when

`a = 17, b = 5, c = 7, d = 3` and `e = 10`

Show the changing contents of the stack as the RPN expression is evaluated.

[4]

6 The diagram shows a logic circuit.



(a) Complete the truth table for the given logic circuit.

Show your working.

			Working space				
A	**B**	**C**	**P**	**Q**	**R**	**S**	**Z**
0	0	0					
0	0	1					
0	1	0					
0	1	1					
1	0	0					
1	0	1					
1	1	0					
1	1	1					

[3]

(b) Write the Boolean expression that corresponds to the logic circuit as a sum-of-products.

Z = ...

....

....

.... [2]

(c) **(i)** Complete the Karnaugh map (K-map) for the Boolean expression:

$\overline{A}.\overline{B}.\overline{C} + \overline{A}.\overline{B}.C + A.\overline{B}.\overline{C} + A.\overline{B}.C + A.B.\overline{C} + A.B.C$

A \ BC	00	01	11	10
0				
1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to part **(c)(ii)** as a simplified sum-of-products.

....

.... [1]

7 **(a)** Describe what is meant by a digital certificate.

....

....

....

....

....

.... [3]

(b) Explain the role of a digital certificate in creating a digital signature.

....

....

....

.... [2]

(c) Describe the performance of a binary search in relation to the number of data items in the array being searched. Refer to Big O notation in your answer.

....

....

....

.... [2]

11 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

(a) State **two** features of RISC processors.

....

....

....

.... [2]

(b) Outline the process of interrupt handling as it could be applied to RISC or CISC processors.

....

....

....

....

....

.... [3]

(c) Explain how pipelining affects interrupt handling for RISC processors.

....

....

....

....

....

.... [3]

5 **(a)** Write this infix expression in Reverse Polish Notation (RPN):

`(7 − 2 + 8) / (9 − 5)`

....

....

....

.... [2]

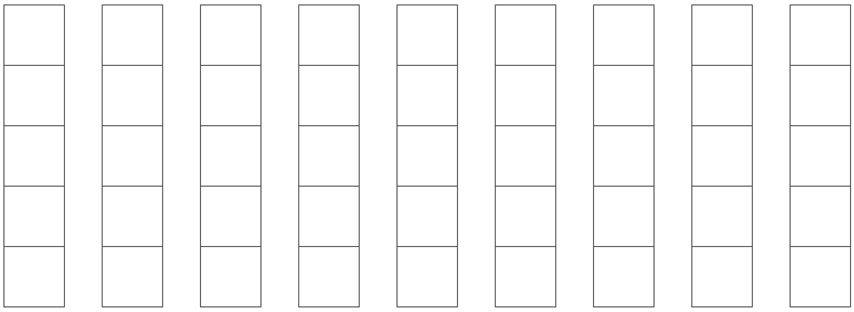
(b) Evaluate this RPN expression:

`a d + a b + c − *`

when

`a = 6`, `b = 3`, `c = 7` and `d = 9`

Show the changing contents of the stack as the RPN expression is evaluated.



[4]

(c) Write this RPN expression in infix form:

`b a c − + d b + * c /`

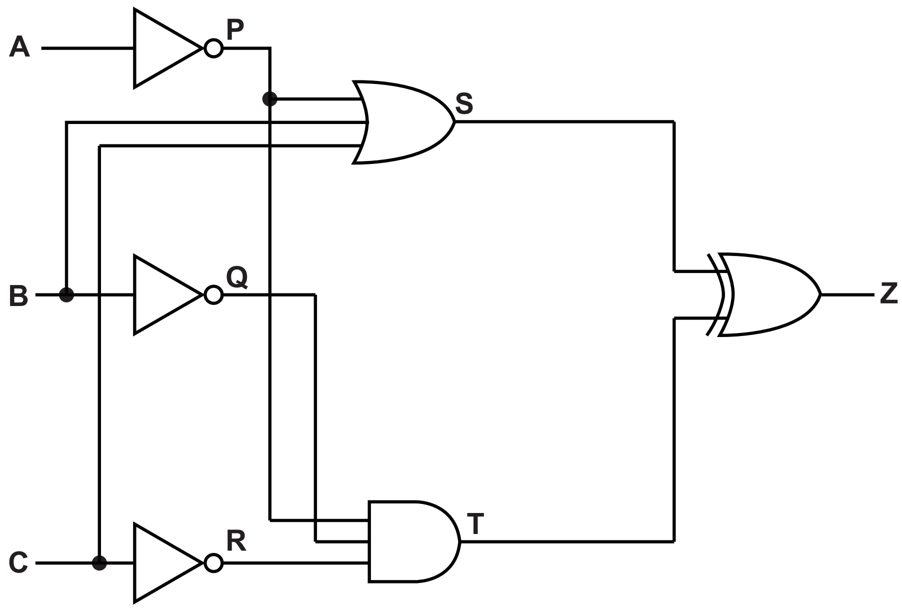
....

....

....

.... [3]

6 The diagram shows a logic circuit.



(a) Complete the truth table for the given logic circuit.
Show your working.

			Working space					
A	**B**	**C**	**P**	**Q**	**R**	**S**	**T**	**Z**
0	0	0						
0	0	1						
0	1	0						
0	1	1						
1	0	0						
1	0	1						
1	1	0						
1	1	1						

[3]

(b) Write the Boolean expression that corresponds to the logic circuit as a sum-of-products.

Z = ...

....

....

.... [2]

(c) **(i)** Complete the Karnaugh map (K-map) for this Boolean expression:

$$\overline{A}.\overline{B}.\overline{C} + \overline{A}.B.\overline{C} + \overline{A}.B.C + A.\overline{B}.\overline{C} + A.B.\overline{C} + A.B.C$$

A \ BC	00	01	11	10
0				
1				

[2]

(ii) Draw loop(s) around appropriate group(s) in the K-map to produce an optimal sum-of-products. [2]

(iii) Write the Boolean expression from your answer to part **c(ii)** as a simplified sum-of-products.

....

.... [1]

7 **(a)** Outline what is meant by **direct access** as a method of file access.

....

....

....

.... [2]

(b) Explain how direct access is used to locate a specific record in sequential files and random files.

(i) Sequential files ....

....

....

.... [2]

(ii) Random files ....

....

....

.... [2]